

Ideation Phase

Define the Problem Statements

Date	13 July 2026
Team ID	[To be filled by student/team]
Project Name	Voice-Based Concept Understanding Analyzer (VBCUA)
Maximum Marks	2 Marks

Customer Problem Statement Template

The customer problem statement is used to understand the challenges faced by the intended users of the Voice-Based Concept Understanding Analyzer (VBCUA) from their own point of view. It focuses on what matters most to students, self-learners, and evaluators when assessing spoken conceptual understanding, and helps the project team design a solution that genuinely addresses their needs.

A well-articulated customer problem statement enables the team to identify the most appropriate solution for the challenges faced by users of the application. It also builds empathy for the customer, providing a clearer understanding of how they perceive the process of preparing, delivering, and evaluating spoken explanations of academic concepts.

Reference: <https://miro.com/templates/customer-problem-statement/>

Problem Statements

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a student preparing spoken explanations of academic concepts for assessment	clearly communicate my understanding of a concept and know whether my explanation is actually correct and complete	I have no quick, objective way to check if my spoken answer covers the key ideas before it is graded	manual review of spoken answers is slow, and self-assessment of one's own explanation is difficult without a reference to compare against	anxious and unsure about how well I actually understand the topic
PS-2	a teacher or evaluator assessing multiple students' spoken explanations of a concept	evaluate each student's conceptual understanding consistently, fairly, and efficiently	manually listening to, transcribing, and comparing every audio response against a model answer is extremely time-consuming and prone to inconsistent grading	there is no automated tool that transcribes spoken responses and objectively measures their semantic closeness to a reference answer	overwhelmed by the grading workload and concerned about fairness across students
PS-3	a self-directed learner practising verbal explanations of	get immediate, structured feedback on my spoken	I do not have access to an instructor at the moment I want	traditional self-study tools focus on text-based quizzes rather than	discouraged from practising verbally, since I

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	concepts outside a classroom	explanation so I can identify gaps in my understanding	feedback, and typing out my explanation defeats the purpose of practising verbal recall	evaluating spoken conceptual explanations	cannot verify my own progress